Aravinda Boovaraghavan

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Education

Carnegie Mellon University Pittsburgh, PA
Master of Information Systems Management August 2024 - December 2025
Coursework: ML in Production, Intro to ML, Generative AI, Distributed Systems, Data Structures & Algorithms, Databases

Vellore Institute of Technology Chennai, India
Bachelor of Technology - Computer Science and Engineering June 2019 - July 2023

Skills

Programming & Frontend: Java, Python, JavaScript, TypeScript, SQL, C++, R, ReactJS, HTML, CSS, Redux

Backend & APIs: REST APIs, GraphQL, Spring Boot, Flask, FastAPI, Microservices, Apache Kafka, Spark, Unit Testing

AI/ML: AI Agents, PyTorch, TensorFlow, scikit-learn, LangChain, LLMs (GPT, Claude), RAG, NLP, Prompt Engineering

Data & Analytics: Pandas, NumPy, DuckDB, Grafana, Tableau, Data Modeling

Databases: PostgreSQL, MongoDB, Vector DBs (Pinecone, Chroma)

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Git/GitHub, CI/CD, Cloud-Native Architecture

Research Experience

Carnegie Mellon University Pittsburgh, PA
Research Assistant - Professor Mohsen Foroughifar January 2026 - June 2026

- Engineered large-scale ETL pipelines in Python over 9M+ ad auction records stored in Parquet format, using DuckDB and Apache Spark to ensure data quality, consistency, and high-performance distributed processing.
- Designed modular, idempotent data workflows with monitoring and data quality checks to compute per-bidder performance metrics and surface revenue impact patterns across distributed transaction logs at scale.

Research Assistant - Professor Rahul Telang May 2025 - October 2025

- Designed and implemented a scalable data ingestion and entity-resolution pipeline over 1M+ U.S. copyright registrations using Python and SQL; developed fuzzy-matching algorithms and web scraping workflows to link external records to official Copyright Office data, achieving a 97% match rate across a 43-year dataset.

Experience

MCFA Global Philadelphia, PA
AI Systems Management Intern June 2026 - Present

- Designed a scalable three-layer AI-driven federal contract intelligence pipeline to shift from reactive RFP discovery to proactive opportunity forecasting, surfacing leads 3–18 months ahead of public posting.
- Built Python data ingestion scripts integrating SAM.gov and USASpending.gov REST APIs, resolving production constraints including rate limiting, award type code groupings, and pagination to retrieve and filter live federal contract data.

PricewaterhouseCoopers (PwC) India Chennai, India
Associate Technology Consultant - Emerging Technologies July 2023 - April 2024

- **Generative AI:** Built a production-grade document Q&A platform with a Flask REST API backend, integrating GPT-3, Llama 2, and Azure services for natural language querying over enterprise documents; fine-tuned models on domain-specific data to improve answer accuracy and reduce manual document review time.
- **Backend Development:** Designed and developed scalable backend services and RESTful/GraphQL APIs (Spring Boot, Java) in a microservices architecture, translating ambiguous stakeholder requirements into technical trade-off decisions that balanced usability and system reliability for thousands of users across a state government ranking portal.
- **Deployment & Workflows:** Built real-time communication features and admin workflows to support data validation and reporting across public, official, and administrative user tiers; deployed on Kubernetes for containerized production serving.

Technology Consultant Intern - Emerging Technologies January 2023 - July 2023

- **Full-Stack Development:** Built and integrated frontend components and backend API connections for a state government institutional ranking portal using ReactJS and RESTful APIs, implementing role-based views for public, official, and administrative users with Chart.js visualizations.

Projects

AI-Powered Movie Recommendation System [Machine Learning, MLOps] August 2025 - December 2025

- Developed a large-scale movie recommendation system serving 1M users and 27K movies by training and evaluating matrix factorization models (SVD, ALS) on 2.5M user–movie interactions derived from Kafka logs.
- Engineered a scalable, fault-tolerant production framework integrating Flask, Docker, and CI/CD pipelines, with drift monitoring, A/B testing, and versioning for model, data, and pipelines; validated system reliability through load testing across 2,000 requests achieving 100% success rate at 115.94ms average latency.

Multimodal Product Intelligence System [Diffusion Models, RAG] October 2025 - December 2025

- Built an orchestrated multi-agent pipeline to extract structured product attributes from Amazon reviews and descriptions using a retrieval-augmented workflow with BERT embeddings and FAISS indexing.